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$$\begin{aligned} & \lim_{x \rightarrow \infty} \left((x-7) \tan \left(\frac{-1}{2x+3} \right) \right) \\ &= \lim_{x \rightarrow \infty} (x-7) \cdot \lim_{x \rightarrow \infty} \left(\frac{\sin \left(\frac{-1}{2x+3} \right)}{\cos \left(\frac{-1}{2x+3} \right)} \right) \cdot \lim_{x \rightarrow \infty} \left(\frac{\frac{-1}{2x+3}}{\frac{-1}{2x+3}} \right) \\ &= \lim_{x \rightarrow \infty} (x-7) \left(\frac{-1}{2x+3} \right) \cdot \lim_{x \rightarrow \infty} \left(\frac{\sin \left(\frac{-1}{2x+3} \right)}{\frac{-1}{2x+3}} \right) \cdot \lim_{x \rightarrow \infty} \left(\frac{1}{\cos \left(\frac{-1}{2x+3} \right)} \right) \\ &= \lim_{x \rightarrow \infty} \frac{-x+7}{2x+3} \\ &= \lim_{x \rightarrow \infty} \frac{-x}{2x} = \frac{-1}{2} \end{aligned}$$

References